

BajaCLIP

Where each RNA-binding protein leaves its footprint — from sequence alone.

A sequence model that scores intrinsic protein–RNA binding potential across a transcript, for 170 RNA-binding proteins, with no reference genome required.

170

RBPs scored per window

48

reliable RBPs · AUROC $\geq$ 0.90

64 nt

sliding window

Trained on ENCODE eCLIP + POSTAR3 · bundled weights (~7.6 MB), no download, no reference genome

Knowing which RNA-binding protein (RBP) touches a transcript, and where, is the first question in almost every post-transcriptional study — splicing, stability, localization, translation.

Experimental footprinting (eCLIP) answers it one protein and one cell type at a time. **BajaCLIP** answers the sequence-intrinsic version of the question directly: given a transcript, where is each RBP most likely to bind?

A compact sphere-CNN reads a 64-nt window and emits a sigmoid binding probability for each of 170 RBPs. Sliding that window along a sequence yields a per-position binding profile for any chosen protein — TARDBP (TDP-43), for example — that drops straight onto the same track / sashimi layers as the rest of the Baja stack. It is **binding potential** — where to look — learned from sequence, strongest at the region level, and most trustworthy for the proteins whose motifs the data actually pin down.

01 · WHAT IT PREDICTS

A per-position footprint, one channel per protein

The model is a multi-label classifier: one 64-nt window in, 170 probabilities out — one per RBP. To profile a transcript you slide the window (default step 8 nt) and read the channel for the protein you care about, giving a smooth per-position track of binding potential. Because every protein is a separate output channel of the same network, scoring all 170 at a position costs a single forward pass.

The training signal is sequence-intrinsic. BajaCLIP learns the motifs and local context that make a window bindable, not the cell-type-specific occupancy that depends on expression and competition. That is deliberate: it answers “where could this protein bind on this sequence,” which is exactly the prior a browser overlay or a cryptic-site screen needs.

02 · PERFORMANCE

Strong where the data is strong: the reliable RBPs

Held-out performance varies by protein, so BajaCLIP ships an explicit reliability list rather than a single headline number. **48 RBPs clear held-out AUROC $\geq$ 0.90** (range 0.901–0.975); these are the proteins whose footprints the model resolves most cleanly, and the ones a profile should be trusted on first. A sample of the top of that list:

RBP	Held-out AUROC	eCLIP positives	Note
FKBP4	0.975	204	
ELAC2	0.972	918	
NKRF	0.966	596	
TARDBP (TDP-43)	0.960	993	ALS/FTD cryptic-exon factor
XRN2	0.958	263	
GEMIN5	0.953	500	
DDX3X	0.952	1,079	
EWSR1	0.947	1,009	

48 of 170 RBPs reach AUROC ≥ 0.90 on held-out data; the full reliability table (RBP, AUROC, positives, operating thresholds) ships with the model.

03 · HONEST SCOPE

What the score is — and what it is not

- **Binding potential, not occupancy.** BajaCLIP scores what the sequence permits, independent of cell type. Whether a protein is actually bound in your sample also depends on its expression, competition and localization — feed in cell-matched context to sharpen it.
- **Region-level, not base-precise.** The footprint is trustworthy as a local profile — which stretch of transcript a protein prefers — rather than as a single-nucleotide crosslink call.
- **Specificity is real, but GC is a confound.** On a held-out bound-vs-other-RBPs test — where every window is a genuine binding site so only protein identity can help — the model separates the correct RBP from the wrong one decisively. But bound windows are also GC-rich, and GC content alone recovers much of a naive bound-vs-background score; the reliability list is built to reward true specificity, not GC.
- **Binding is not regulation.** Predicting where a protein binds does not predict which exons move when that protein is removed — knockdown effects are largely indirect. BajaCLIP is a map of footprints, not a causal model of regulation.

04 · SOFTWARE

Runs from sequence, ships with its weights

BajaCLIP is packaged like BajaSplice: the trained checkpoint (bajacclip_predict.v1, ~7.6 MB) is bundled, so there is no download step and no reference genome to stage. Point it at a sequence, choose `all` / `reliable` / a comma list of RBPs, and read the profile.

```
from bajacclip.scan import load_model, scan_sequence, resolve_rbps
m = load_model() # bundled checkpoint
names, cols = resolve_rbps(m, 'TARDBP') # 'all' | 'reliable' | list
centers, scores = scan_sequence(m, seq, cols, step=8)
# centers: window-centre positions · scores: (n_windows, len(cols))
```

SUMMARY

A footprint map for 170 RNA-binding proteins, from sequence.

BajaCLIP turns a transcript into a per-position binding profile for any of 170 RBPs with a single sphere-CNN — no genome, no download. It is strongest for the 48 reliable proteins (held-out AUROC ≥ 0.90), it reports binding potential rather than cell-specific occupancy, and it is honest about its limits: specificity is real but GC-confounded, and binding is a map, not a causal model of regulation. Use it to decide where to look.